

GHARDAIA NOUMERAT, Algeria

WMO: 605660

Lat: 32.384N

Lon: 3.794E

Elev: 1512

StdP: 13.91

Time Zone: 1.00 (E01)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	39.1	41.1	14.4	12.1	58.1	18.0	14.4	60.3	32.4	53.2	28.5	53.5	5.5	330	0.492

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.4	110.9	69.8	108.7	69.3	106.1	68.5	73.4	99.9	71.7	98.7	70.5	97.4	9.8	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.2	102.3	80.1	64.1	94.6	80.5	61.9	87.4	81.3	37.8	99.4	36.2	98.1	35.1	97.7	82.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.1	23.2	20.4	35.2	113.6	2.1	1.6	33.7	114.8	32.5	115.7	31.4	116.6	29.9	117.8
			30.5	76.0	2.0	3.1	29.1	78.3	27.9	80.1	26.8	81.9	25.4	84.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	72.6	53.1	55.9	62.8	69.8	79.0	87.9	94.5	92.8	84.0	73.9	61.8	54.6	(6)
(7)		DBStd	15.54	4.83	5.74	6.54	6.53	6.63	5.87	4.67	4.52	6.41	6.09	5.68	4.87	(7)
(8)		HDD50	49	23	11	1	0	0	0	0	0	0	0	1	13	(8)
(9)		HDD65	1232	370	260	121	24	1	0	0	0	0	4	128	324	(9)
(10)		CDD50	8300	119	176	398	594	899	1137	1380	1326	1019	741	355	156	(10)
(11)		CDD65	4009	1	6	52	168	435	687	915	861	570	280	32	2	(11)
(12)		CDH74	58917	4	53	513	1780	5463	10587	15724	14249	7691	2627	216	10	(12)
(13)		CDH80	37320	0	10	133	705	2898	6815	11322	9890	4439	1070	38	0	(13)
(14)	Wind	WSAvg	8.1	7.8	8.8	9.3	10.1	9.7	8.9	7.7	7.3	7.4	6.5	6.8	7.5	(14)
(15)	Precipitation	PrecAvg	2.3	0.3	0.1	0.4	0.2	0.1	0.1	0.0	0.1	0.2	0.2	0.3	0.2	(15)
(16)		PrecMax	5.2	2.6	0.9	2.2	2.1	0.8	0.7	0.7	0.8	1.7	1.1	1.2	1.6	(16)
(17)		PrecMin	0.7	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(17)
(18)		PrecStd	0.9	0.5	0.2	0.5	0.4	0.2	0.2	0.1	0.2	0.4	0.3	0.3	0.4	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	73.5	79.6	89.4	96.5	105.1	109.9	114.0	111.9	106.8	98.3	84.3	75.1	(19)
(20)		0.4%	MCWB	53.1	56.2	59.9	63.2	65.7	69.3	70.8	70.5	68.0	65.9	60.0	56.5	(20)
(21)		2%	DB	69.5	74.0	83.7	91.6	100.5	107.6	111.5	109.6	103.4	93.5	79.6	70.2	(21)
(22)		2%	MCWB	52.1	53.3	57.2	60.2	65.0	68.6	69.9	69.8	67.2	63.6	58.6	53.9	(22)
(23)		5%	DB	66.0	70.5	79.2	87.7	96.9	104.9	109.6	107.5	100.3	89.6	75.7	67.0	(23)
(24)		5%	MCWB	50.5	52.2	55.8	59.1	63.4	67.4	69.3	69.4	67.0	62.9	57.4	52.5	(24)
(25)		10%	DB	62.8	67.2	75.7	84.0	93.2	101.9	107.5	105.1	97.0	85.9	72.6	64.2	(25)
(26)		10%	MCWB	49.0	50.8	54.3	58.4	62.6	66.5	69.0	68.7	66.9	62.5	56.2	51.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	56.1	59.7	62.4	65.7	69.2	73.7	75.5	76.6	73.1	70.1	64.9	58.9	(27)
(28)		0.4%	MCDWB	68.6	73.6	78.6	85.2	94.9	101.5	103.9	103.3	91.9	83.6	77.7	69.9	(28)
(29)		2%	WB	53.3	55.7	60.0	62.9	66.6	70.8	73.0	73.5	71.1	67.7	61.9	55.9	(29)
(30)		2%	MCDWB	65.7	70.9	78.2	82.0	91.9	99.0	102.7	99.8	90.1	81.2	74.1	66.1	(30)
(31)		5%	WB	51.5	53.2	57.5	61.0	65.3	69.0	71.3	71.7	69.8	66.2	59.4	53.9	(31)
(32)		5%	MCDWB	63.3	67.4	75.4	81.2	90.5	97.8	102.7	98.6	90.1	81.4	72.5	64.3	(32)
(33)		10%	WB	50.0	51.5	55.6	59.4	63.8	67.6	70.1	70.4	68.7	64.8	57.3	52.4	(33)
(34)		10%	MCDWB	61.1	65.2	73.0	79.7	88.7	96.5	101.3	97.5	89.3	80.6	70.2	62.6	(34)
(35)	Mean Daily Temperature Range		MDBR	17.8	18.8	20.4	21.4	21.9	22.3	22.4	21.6	20.5	19.4	18.3	17.4	(35)
(36)		5% DB	MCDDBR	21.9	22.7	24.8	26.1	25.2	25.2	24.1	23.8	23.9	23.4	21.6	20.8	(36)
(37)			MCWBR	11.1	11.5	11.0	10.7	10.0	9.0	8.4	8.7	8.1	8.9	10.0	10.6	(37)
(38)			5% WB	MCDDBR	19.5	20.4	22.5	22.3	22.4	22.6	22.3	21.7	20.8	19.1	19.1	18.2
(39)		MCWBR		10.8	11.3	10.8	10.5	9.7	9.4	9.0	8.5	8.0	8.0	9.9	9.8	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.315	0.345	0.438	0.481	0.537	0.569	0.619	0.584	0.549	0.469	0.370	0.330	(40)	
(41)		taud	2.343	2.241	1.934	1.850	1.740	1.682	1.600	1.677	1.754	1.920	2.178	2.306	(41)	
(42)		Ebn at Noon	285	286	265	259	245	236	224	229	230	239	260	272	(42)	
(43)		Edh at Noon	32	39	57	65	74	78	84	76	68	53	37	31	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1172	1476	1846	2178	2350	2502	2460	2236	1865	1529	1218	1054	(44)	
(45)		RadStd	56	62	78	84	75	55	45	62	64	75	52	52	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	+0.76	N/A	-4.07	N/A	-1.64	-1.12	N/A	N/A	N/A	N/A	(46)
(47)	Regional (2 neighbors)	+0.61	N/A	-4.07	N/A	-1.64	-1.12	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page